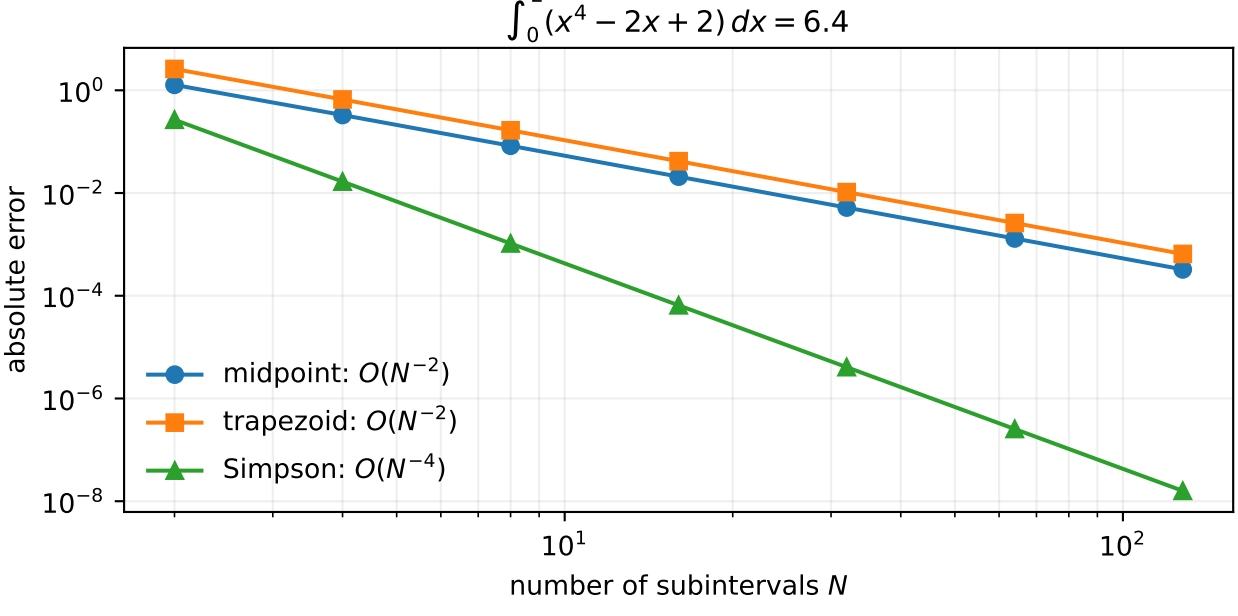
$\int_0^2 (x^4 - 2x + 2)\,dx = 6.4$
absolute error
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
midpoint: $O(N^{-2})$
trapezoid: $O(N^{-2})$
Simpson: $O(N^{-4})$
10^{1}
10^{2}
number of subintervals N